Asmaa Eldesoukey

aeldesou@uci.edu/asmaae@iastate.edu Website Google Scholar

EDUCATION

University of California (UC) Irvine
Ph.D., Mechanical and Aerospace Engineering 2025
Focus: Dynamical and Control Systems
Thesis: *Inference and Collective Steering for Stochastic and Deterministic Dynamics.*

M.S., Mechanical and Aerospace Engineering 2021
Focus: Dynamical and Control Systems

Zewail University of Science and Technology Giza, Egypt
B.Sc., Aerospace Engineering 2018
Thesis: *A Geometric Approach to Modeling, Simulation, and Control of Rotational Rigid Body Motions.*

ACADEMIC POSITIONS

Postdoctoral Research Associate Aerospace Engineering, Iowa State University
March 2026 – Present Ames, IA
Advisor: Prof. Abhishek Halder
RESEARCH: *Developing scalable control methods for interacting multi-agent systems, combining mean-field models, stochastic control, optimization, and data-driven decision-making.*

Postdoctoral Researcher Mechanical and Aerospace Engineering, UC Irvine
July 2025 – Feb. 2026 Irvine, CA
Advisor: Prof. Tryphon T. Georgiou
RESEARCH: *Developed data-driven control methods from partial and aggregate observations, with applications to latent dynamical modeling.*

Graduate Student Researcher Mechanical and Aerospace Engineering, UC Irvine
Fall 2019 – Spring 2025 Irvine, CA
Advisor: Prof. Tryphon T. Georgiou
RESEARCH: *Developed theory and computational methods for stochastic control, optimal transport, and probabilistic inference, with applications to networked systems.*

RESEARCH INTERESTS

- Dynamics, Control, and Optimization
- Autonomous and Multi-Agent Systems
- Stochastic Systems and Estimation

JOURNAL AND CONFERENCE PUBLICATIONS

10. [Journal & Conference] **Eldesoukey A.**, Chen Y., Halder A. (2026). "A Generalized Sinkhorn Algorithm for Mean-Field Schrödinger Bridge." *IEEE Control Systems Letters*, 10, 1087–1092. Also to be presented at the IEEE Conference on Decision and Control (CDC).
9. [Journal & Conference] Sabbagh R., **Eldesoukey A.**, Abdelgalil M., Georgiou T.T. (2026). "On the Isospectral Nature of Minimum-Shear Covariance Control." *IEEE Control Systems Letters*, 10, 1171–1176. Also presented at the IEEE Conference on Decision and Control (CDC).
8. [Journal] Sabbagh R., **Eldesoukey A.**, Abdelgalil M., Georgiou T.T. (2025). "Minimizing Control Attention: The Linear Gauss–Markov Paradigm." *IEEE Transactions on Automatic Control*. Accepted/In press.

7. [Conference] **Eldesoukey A.**, Abdelgalil M., Georgiou T.T. (2025). "Collective Steering: Tracer-Informed Dynamics." *Proceedings of the IEEE Conference on Decision and Control (CDC)*, pp. 3135–3141.
6. [Journal] **Eldesoukey A.**, Miangolarra O.M., Georgiou T.T. (2025). "Inferring Evanescent Stochastic Dynamics from Marginal Distributions." *Journal of Systems Science & Complexity*, 38(1), 129–149.
5. [Journal] Miangolarra O.M., **Eldesoukey A.**, Miangolarra A.M., Georgiou T.T. (2025). "Maximum Entropy Inference of Reaction–Diffusion Models." *The Journal of Chemical Physics*, 162, 194104.
4. [Journal] **Eldesoukey A.**, Georgiou T.T. (2025). "Schrödinger's Control and Estimation Paradigm with Spatio-Temporal Distributions on Graphs." *IEEE Transactions on Automatic Control*, 70, 2466–2478.
3. [Journal] Miangolarra O.M., **Eldesoukey A.**, Georgiou T.T. (2024). "Inferring Potential Landscapes: A Schrödinger Bridge Approach to Maximum Caliber." *Physical Review Research*, 6, 033070.
2. [Journal & Conference] **Eldesoukey A.**, Miangolarra O.M., Georgiou T.T. (2024). "An Excursion onto Schrödinger's Bridges: Stochastic Flows with Spatio-Temporal Marginals." *IEEE Control Systems Letters*, 8, 1138–1143. Also presented at the IEEE Conference on Decision and Control (CDC), 2024.
1. [Conference] Abdelgalil M., **Eldesoukey A.**, Taha H. (2023). "Singularly Perturbed Averaging with Application to Bio-Inspired 3D Source Seeking." *Proceedings of the American Control Conference (ACC)*, pp. 885–890.

BOOKS

Eldesoukey A. (2026). "Inference and Collective Steering for Stochastic and Deterministic Systems: New Directions in Schrödinger Bridges and Optimal Transport." *Springer Nature*, In Production.

PREPRINTS

3. **Eldesoukey A.**, Haider M.Z., Napolitano I., Chen Y., Halder A. (2026). "Second-Order Mean-Field Schrödinger Bridges with Partial State Information." arXiv:2610.02537.
2. **Eldesoukey A.**, Haider M.Z., Napolitano I., Chen Y., Halder A. (2026). "Schrödinger Bridges over Kinetic Swarming Models." arXiv:2608.25281.
1. Bondar G.A., **Eldesoukey A.**, Chen Y., Halder A. (2026). "Nonlinear Non-Gaussian Density Steering with Input and Noise Channel Mismatch: Sinkhorn with Memory for Solving the Control-Affine Schrödinger Bridge Problem." arXiv:2604.23370.

HONORS AND AWARDS

- Springer Theses Award, Recognizing Outstanding Ph.D. Research 2026
- Outstanding Student Paper Award, IEEE Control Systems Society 2024
- Paul and Beverly Holmes Endowed Fellowship, UC Irvine 2024
- Best Student Paper Award, American Control Conference (ACC) 2023
- Division of Teaching Excellence and Innovation Fellowship, UC Irvine 2020
- Magna Cum Laude Honors, Zewail UST 2018
- Provost Honor Roll, Zewail UST 2013–2018

TEACHING EXPERIENCE

Graduate Teaching Assistant — UC Irvine
Winter 2020 – Spring 2024 — Irvine, CA

Courses:

- Introduction to Control Systems (Fall 2020, Winter 2024)
- Aerospace Laboratory (Spring 2021, Fall 2021–2022, Spring 2023–2024)
- Machine Design (Winter 2023)
- Engineering Thermodynamics (Fall 2023)
- Viscous Compressible Flows (Spring 2022)
- Dynamics of Aerospace Vehicles (Winter 2020–2022)
- Introduction to Fluid Mechanics (Summer 2021)

Teaching Responsibilities:

- Led weekly tutorial and laboratory sessions of 1–3 hours with up to 80 students.
- Provided individualized student support.
- Prepared course assessments, including exams and homework.
- Designed and supervised course projects.
- Completed pedagogy certification and developed course materials and digital resources across multiple learning-management systems.

Graduate Teaching Assistant — Zewail UST
Fall 2018 – Spring 2019 — Giza, Egypt

Courses:

- Senior Design Projects (Fall 2018, Spring 2019)
- Nonlinear Control Systems (Spring 2019)
- Attitude Determination and Control (Spring 2019)
- Advanced Control Systems (Fall 2018)
- Engineering Dynamics (Fall 2018)

Teaching Responsibilities:

- Mentored senior undergraduate students in capstone design projects.
- Led weekly 3-hour tutorial and laboratory sessions with up to 30 students.
- Prepared course assessments, including written exams and problem sets.
- Integrated control and robotics software (ROS & Gazebo) into coursework.

COAUTHORED PRESENTATIONS

- IEEE Conference on Decision and Control, Rio de Janeiro, Brazil — Dec. 2025
  "Collective Steering: Tracer-Informed Dynamics."
- American Physical Society March Meeting, Anaheim, California — Mar. 2025
  "Maximum-Entropy Inference of Reaction–Diffusion Models."
- IEEE Conference on Decision and Control, Milan, Italy — Dec. 2024
  "An Excursion onto Schrödinger's Bridges: Stochastic Flows with Spatio-Temporal Marginals."
- 26th International Symposium on Mathematical Theory of Networks and Systems, Cambridge, UK — Aug. 2024
  "Steering Stochastic Flows on Graphs between Marginals in Space and Time."
- 43rd SoCal Control Workshop, UC Los Angeles — Apr. 2024
  "Steering Stochastic Flows under Soft Spatio-Temporal Conditioning."

- American Physical Society March Meeting, Minneapolis, Minnesota Mar. 2024
  "Partially Observed Schrödinger Flows: An Application to Stochastic Thermodynamics."
- Workshop on Schur Analysis and Applications to Hypercomplex Analysis, Neural Networks and Linear Systems, Chapman University Jan. 2024
  "Schrödinger Bridges between Marginal Distributions in Space and Time on Graphs."
- 42nd SoCal Control Workshop, UC Riverside Nov. 2023
  "Schrödinger's Control and Estimation Paradigm with Stopping Times on Graphs."

SERVICE AND OUTREACH

Reviewer
Current
IEEE Transactions on Automatic Control
IEEE Control Systems Letters
IEEE Transactions on Control of Network Systems
IEEE Conference on Decision and Control
Systems & Control Letters
Latin American Journal of Probability and Mathematical Statistics

Graduate Mentor Society of Women Engineers
Jan. 2025 – Jan. 2026 UC Irvine
ROLE: *Mentored a newly admitted doctoral student on program requirements, research skill development, and professional development planning.*

Member Graduate Student Association
Sept. 2021 – Jan. 2022 UC Irvine
ROLE: *Organized professional development workshops for graduate students in Mechanical and Aerospace Engineering.*

Student Mentor Robotics Lab, Zewail UST
Sept. 2014 – Sept. 2017 Giza, Egypt
ROLE: *Organized and conducted workshops for undergraduate engineering students on embedded systems, microcontroller programming, and CubeSat mission design.*

ADDITIONAL PROFESSIONAL EXPERIENCE

Control Laboratory Engineer and Co-founder Zewail UST
June 2018 – June 2019 Giza, Egypt
TECHNICAL ROLE: *Designed experimental platforms for control, robotics, and dynamical-systems instruction, including cart–pendulum and robotic-arm setups.*
ADMINISTRATIVE ROLE: *Defined equipment specifications, procurement workflows, and laboratory infrastructure supporting experimental research and hands-on instruction.*

Summer Intern IBM
Summer 2016 Smart Village, Egypt
PROJECT: Monitoring and analyzing driver violations from a data-acquisition system using IBM Watson (in collaboration with DrivePal).
ROLE: *Developed software for depth-data acquisition from dual-camera systems, enabling automated detection of driver violations for real-time monitoring applications.*